Machine Learning Models for Predicting Agricultural Yields in Developing Countries

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1. Introduction

The food security in developing countries is also shallow because of uncertain production. Climatic change, including shifts in rainfall, temperature, and soil quality, renders conventional methods used in prediction unreliable (Oishy et al., 2025). The farmers and policymakers lack the tools for accurate predictions, poor use of resources results in financial losses, and they are prone to crises.

Accordingly, proper prediction of yields is crucial for the sustainability of the food chain (Md Abu Jabed and Azmi, 2024). Nonlinear correlations between climate, soil, and crop growth variables are identifiable using machine learning (ML) methods and have a better predictive capacity, which is higher than that of conventional models. Through enhancing the quality of forecasting, this project will be important to climate resilience and effective agricultural planning, as well as sustainability in the long term (Baraj et al., 2024).

Whether they are small-hold farmers, agricultural ministries, NGOs, or policymakers, they are all beneficiaries. Governments can help through the maximization of planting decisions by farmers, as well as better food security plans by farmers (Viana et al., 2021). In the end, improved forecasts of cultivation become the support of the livelihood of millions and aid in stabilizing unstable economies of developing nations.

2. Background

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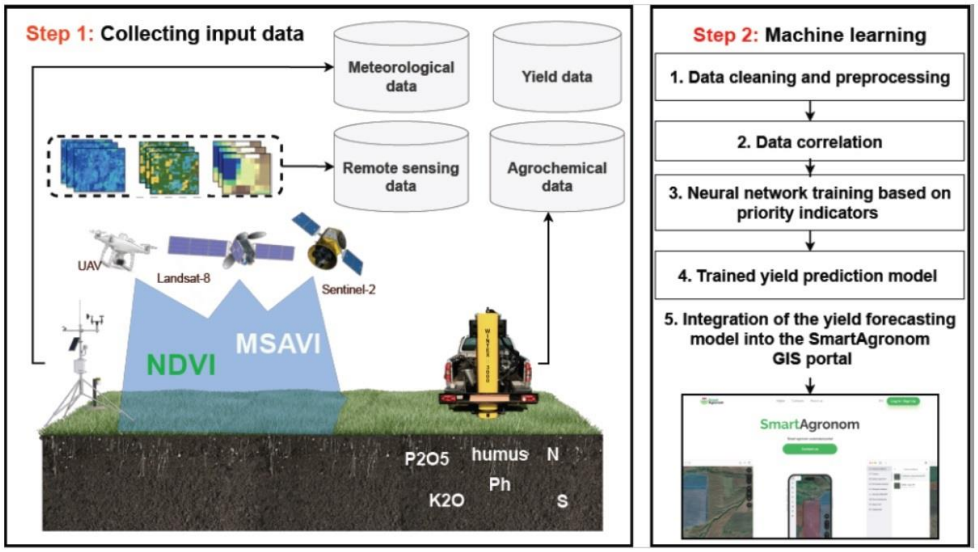
Step 1: Collecting input data
Meteorological data
Yield data
Remote sensing data
Agrochemical data
UAV
Landsat-8
Sentinel-2
NDVI
MSAVI
P2O5
humus
N
Ph
K2O
S
Step 2: Machine learning
1. Data cleaning and preprocessing
2. Data correlation
3. Neural network training based on priority indicators
4. Trained yield prediction model
5. Integration of the yield forecasting model into the SmartAgronom GIS portal
SmartAgronom

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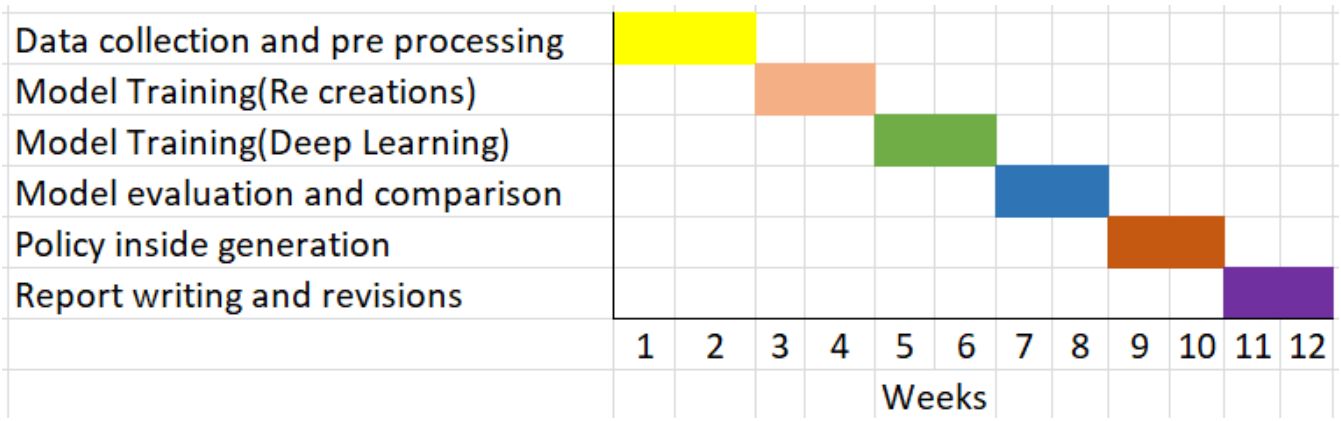

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